



Stay Rational, Resist Illusions: De-biasing Source-Free Domain Adaptation with CLIP-Verified Pseudo-Source Domain

Qing Tian^{1,2,3} · Yongjiang Liu¹ · Keyang Cheng⁴ · Weihua Ou⁵ · Jianping Gou⁶

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Abstract

Source-free domain adaptation (SFDA) aims to adapt a source model pre-trained on a labeled source domain to an unlabeled target domain, under the constraint that the original source data is no longer accessible. While some existing approaches achieve promising adaptation performance by approximating certain target samples as source-like samples to construct a pseudo-source domain and reduce domain discrepancies, they typically rely solely on the source model for this construction. This overlooks a critical issue: the source model may suffer from illusions when identifying pseudo-source samples, due to its inherited spurious correlations between domain-specific features (e.g., environment) and domain-invariant features (e.g., class identity) learned from the source domain. As a result, the reliability of the generated pseudo-source domain is compromised. In this work, we propose a novel Staying Rational and Resisting Illusions (SRRI) method for SFDA to resolve the two fundamental challenges: confirmation bias and domain shift. In particular, we innovatively leverage the CLIP (Contrastive Language-Image Pre-training) model as a source of external knowledge to mitigate feature confounding. SRRI employs knowledge distillation to guide the source model in disentangling class-discriminative causal features from domain-specific spurious features, building upon this, we design a CLIP-guided dual-model validation and class balancing strategy to ensure the reliability and richness of the pseudo-source domain samples even when poor distillation occurs in some tasks. Furthermore, we propose a dynamic pseudo-source domain optimization mechanism that continuously fine-tunes the task-specific prompts of the CLIP during adaptation to further correct the bias of the target model on hard samples, while periodically reconstructing and refining the pseudo-source domain. Ultimately, we apply robust supervised learning and distribution alignment to these respective domains. Extensive experiments show that SRRI outperforms the state-of-the-art methods.

Keywords Source-free domain adaptation · Pseudo-source domain · Knowledge distillation · CLIP-guided

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Extended author information available on the last page of the article



1 Introduction

As a subfield of transfer learning, Unsupervised Domain Adaptation (UDA) relies on access to labeled source domain data and unlabeled target domain data (Pan & Yang, 2010). However, in certain scenarios, accessing source data may be infeasible due to concerns such as privacy and security (Lao et al., 2021; Liang et al., 2019). To address this limitation, Source-Free Domain Adaptation (SFDA) (Kim et al., 2021; Liang et al., 2020) has emerged as a widely studied alternative. Its objective is to enable knowledge transfer using only a pre-trained source model and unlabeled target domain samples, without requiring access to the original source data.

Current SFDA approaches primarily focus on generating or extracting auxiliary information to facilitate unsupervised adaptation. These methods can be broadly categorized into two types. The first seeks to uncover additional supervisory signals from the source model (Huang et al., 2021; Lao et al., 2021; Wang et al., 2022) and target data (Tang et al., 2022; Yang et al., 2021, 2022a), while the second aims to construct a pseudo-source domain (Cui et al., 2023; Du et al., 2024) to simulate a conventional UDA setting. This work centers on the latter approach, exploring how to effectively build a pseudo-source domain to enhance adaptation performance.

As previously mentioned, the source model in SFDA is trained on labeled data from the source domain. However, in most cases, individual samples consist of both domain-invariant features (e.g., category identity) and domain-specific features (e.g., background or environment). When spurious correlations (Li et al., 2025; Yue et al., 2023) exist between these two types of features in the source domain, they can severely impair the generalization ability of the model. As illustrated in Fig. 1, the source model, inheriting knowledge from the source domain, may form misleading associations, mistaking domain-specific features for indicators of certain categories. This illusion causes the model to associate target domain samples with incorrect class identities, leading to the construction of a low-quality pseudo-source domain. To address this issue, we propose a novel Staying Rational and Resisting Illusions (SRRI) method for the source model. Specifically, we leverage the powerful prior knowledge of the pre-trained vision-language model CLIP (Radford et al., 2021). We first distill the rich semantic understanding of the CLIP into the source model to help it overcome misleading correlations. Then, using the updated source model, we generate

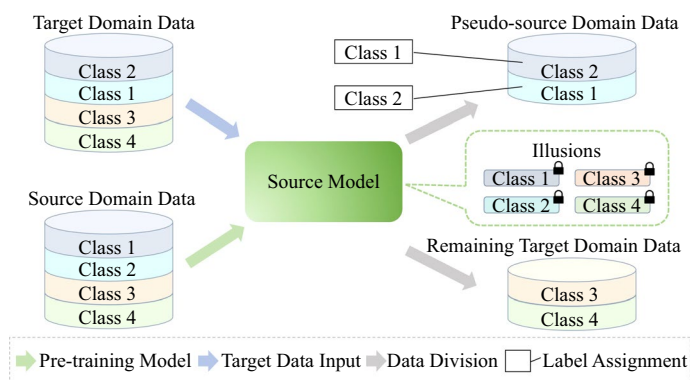


Fig. 1 The effect of source modeling illusions on the quality of pseudo-source domain. The different colors of the cylinders represent distinct domain-specific features

a pseudo-source domain with the assistance of CLIP-based validation and class balancing strategy. Finally, adaptation is performed using a UDA framework composed of three key modules. Importantly, during adaptation, we keep the CLIP model used for validation frozen and instead fine-tune the task-specific prompts of a CLIP to provide guidance for the target model. This decouples pseudo-source domain validation from knowledge guidance and enables the periodic refinement of the pseudo-source domain. Our contributions can be summarized as follows:

1. An innovative Staying Rational and Resisting Illusions (SRRI) approach is proposed that distills the deconfounded knowledge of the CLIP model to mitigate spurious correlations learned by the source model and effectively addresses the unreliability of pseudo-source domains due to confirmation bias in Source-Free Domain Adaptation (SFDA).
2. The CLIP-guided dual-model validation and class balancing strategy in our SRRI approach ensures that high-quality pseudo-source domains can be obtained even in the presence of poor distillation, thus providing absolutely excellent domain environments for subsequent distribution alignment to cope with the challenge of domain bias.
3. We propose a dynamic pseudo-source domain optimization mechanism that fine-tunes the task-specific prompts of a CLIP to provide a more robust teacher signal, thereby improving the model perception of hard samples and periodically refining the quality of the pseudo-source domain.
4. Extensive experiments conducted on standard SFDA benchmarks demonstrate that SRRI substantially improves the quality of the pseudo-source domain and significantly outperforms previous state-of-the-art methods.

The rest of the paper is organized as follows: Sect. 2 briefly reviews related work. In Sect. 3, we describe the proposed SRRI method in detail. Section 4 provides an experimental evaluation of the proposed method. Finally, in Sect. 5, we summarize the contents of this paper and give future research directions.

2 Related Work

In this section, we review the recent source-free domain adaptation methods related to our approach as well as methods for multimodal foundation models.

2.1 Source-Free Domain Adaptation

Existing Source-free Domain Adaptation (SFDA) approaches fall into two distinct categories. The first category comprises pseudo-labeling based self-training methods. These approaches generate pseudo-labels for target domain samples using the source model and leverage them as supervisory signals to guide adaptation. SHOT (Liang et al., 2020) is a pioneering work in this direction, optimizing the model through information maximization and feature clustering. Many subsequent studies have focused on improving pseudo-labels quality by incorporating techniques such as nearest-neighbor clustering (Liang et al., 2022; Yang et al., 2021), active learning strategies (Li et al., 2022b), or designing more robust clustering objectives (Qu et al., 2022). In addition, to further enhance the robustness of

pseudo-labeling, some researchers (Tian et al., 2023; Xu et al., 2025) have begun to explore the structured information between samples. The second category involves source information generation or reconstruction to facilitate adaptation. Some methods employ generative adversarial networks (GANs) to synthesize “pseudo-source” data (Du et al., 2024; Kim et al., 2021; Tian et al., 2022), while others transfer statistical information, such as batch normalization parameters, encoded in the source model to align feature distributions across domains (Ishii & Sugiyama, 2021; Yang et al., 2022b). However, a fundamental limitation shared by these methods is their complete reliance on the biased source model for predictions. Specifically, when there exists a significant distribution shift or spurious correlation between the source and target domains, the source model, distorted by these illusions, can easily fall into a vicious cycle of confirmation bias, leading to cumulative prediction errors (Arazo et al., 2020; Zhang et al., 2021a). This paper focuses on a detailed investigation of the pseudo-source domain quality issue within the second category of approaches.

2.2 Multimodal Foundation Model

The core idea behind the multimodal foundation models is to align representations across different modalities, enabling the model to capture deep semantic associations between vision and language. For instance, text-to-image models such as DALL E (Ramesh et al., 2021) and Stable Diffusion (Rombach et al., 2022) can generate highly realistic images based on complex textual descriptions. Other works, like ALIGN (Jia et al., 2021) and FLAVA (Singh et al., 2022), aim to learn a unified multimodal representation space that supports a wide range of vision-language tasks. Among these foundational multimodal models, CLIP (Contrastive Language-Image Pre-training) (Radford et al., 2021) has made a profound impact on the field of computer vision, thanks to its remarkable zero-shot recognition capability and strong domain robustness. A growing body of research has shown that the zero-shot ability of the CLIP allows it to outperform supervised ImageNet-pretrained models when handling out-of-distribution data, stylistic variations, and various types of data corruption (Yuan et al., 2025; Zhang et al., 2021b). Motivated by these strengths, our proposed SRRI leverages the out-of-domain knowledge of the CLIP to correct the in-domain illusions of the source model. Compared to prior methods that rely solely on the source model to construct pseudo-source domains, our approach effectively mitigates knowledge shift and spurious correlations, thereby enabling the construction of a more reliable and higher-quality pseudo-source domain.

3 The Proposed Method

In this section, we first provide a formal description of the SFDA problem. Then, we outline the proposed SRRI approach and present the details of the knowledge distillation, pseudo-source domain generation, and adaptation phases in turn.

3.1 Problem Statement

In traditional Unsupervised Domain Adaptation (UDA) task, the goal is to learn a model using labeled source domain samples and unlabeled target domain samples. The source

domain $D_S = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ and the target domain $D_T = \{(x_i^t)\}_{i=1}^{n_t}$ share the same set of classes $C = \{c_k\}_{k=1}^K$, but their feature distributions differ. In contrast, Source-Free Domain Adaptation (SFDA) does not provide access to the labeled source domain D_S , instead, it offers a pre-trained source model θ_s obtained from source data and the unlabeled target domain samples D_T . Our objective is to learn a target model θ_t that can construct a higher-quality pseudo-source domain. This process involves the use of the pre-trained source model θ_s , the unlabeled target domain data D_T , and a CLIP model θ_v .

3.2 Overview

As illustrated in Fig. 2, the proposed SRRI framework consists of three stages designed to facilitate the distillation of the CLIP model, the construction of a pseudo-source domain, and the final adaptation. Specifically, in the first stage, we perform knowledge distillation by aligning the probability distributions of the CLIP model θ_v and the source model θ_s on the target domain data, resulting in a target model θ_t . The probability distribution of the CLIP model is computed using a predefined set of textual prompts. Unlike prior work (Li et al., 2024) that relies on a single template (e.g., “a photo of a [CLS]”), our method introduces a diverse prompt library to better exploit knowledge of the CLIP model and enhance transferability. We term this initial, pre-adaptation distillation from the generic frozen CLIP cold-start knowledge distillation, since the target model starts identical to the biased source model with no target-specific signal. In the second stage, we adopt a more effective pseudo-source domain generation strategy. Pseudo-source samples are selected based on the consistency scores between the target model θ_t and the CLIP model θ_v . Compared to previous approach (Du et al., 2024) that rely solely on entropy-based criteria, our method mitigates

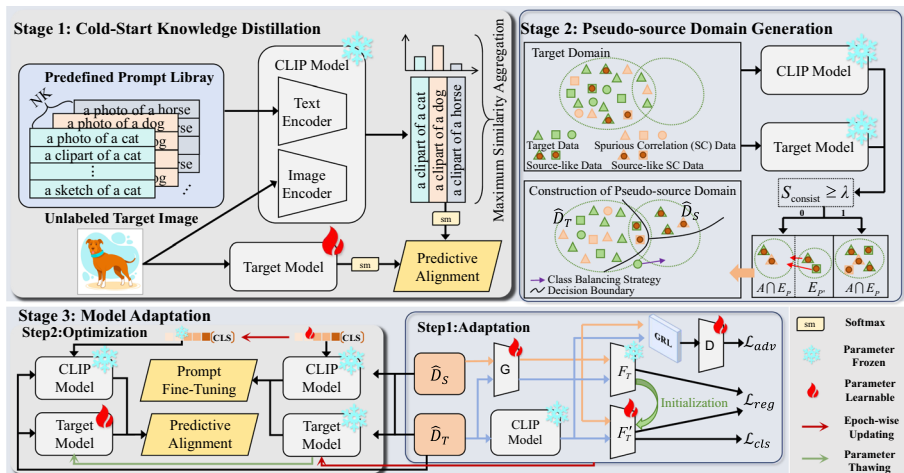


Fig. 2 Overview of our SRRI. The process consists of three stages. Starting from a zero-adaptation and zero target-knowledge setting, we first bootstrap the adaptation process with the general knowledge of the CLIP. We then construct a pseudo-source domain using the proposed dual-model verification and class-balancing strategies. Subsequently, the model is adapted under the standard UDA setting, after which the task-specific prompts of the CLIP are fine-tuned and the refined knowledge is distilled into the target model. The pseudo-source domain is then reconstructed. Except for the initialization stage, these procedures are alternately performed in subsequent rounds

the risk of the target model becoming overconfident in certain predictions. Additionally, we incorporate a class balancing strategy to ensure the robustness of subsequent adaptation. In the third stage, we first perform model adaptation under the standard UDA setting. We then fine-tune the task-specific prompts of a frozen CLIP model and distill the refined CLIP knowledge into the target model on hard samples. Subsequently, we return to Stage 2 to reconstruct the pseudo-source domain. Stage 1 is executed only once, while Stages 2 and 3 are alternately carried out during the remaining adaptation process.

3.3 Cold-Start Knowledge Distillation

3.3.1 Predictive Distributions of Teacher and Student Models

Since the θ_t is initially identical to the heavily biased θ_s , certain forms of knowledge interaction at this stage, such as transferring task-specific knowledge from the target model to CLIP, would inevitably introduce biased supervision. Therefore, what is required during this stage is an objective and unbiased teacher signal to perform cold-start knowledge distillation on the target model, while simultaneously providing a reliable initialization for pseudo-source domain construction. Accordingly, We treat the CLIP model as the teacher and freezing its parameters. To enhance the effectiveness of the distillation process in this stage, we define a prompt library that includes NK predefined templates, such as “a [S] of a [CLS]”, where N represents the number of domain styles, K is the number of classes, S denotes the style descriptor, and CLS is the category embedding. For each sample $x_i^t \in D_T$, we obtain the predicted distribution of the teacher model $q(y | x_i^t)$. For the final prediction score of class c_k , we embed the class name into each prompt template associated with class c_k , and compute their aggregated similarity score with the image as follows:

$$q(y = c_k | x_i^t) = \text{Softmax} \left(\tau \cdot \max_{j=1}^N S(I(x_i^t), T(p_j^k)) \right), \quad (1)$$

where $I(\cdot)$ and $T(\cdot)$ are the image and text encoders of the CLIP model, respectively, p_j^k denotes the j -th prompt for class k , $S(\cdot, \cdot)$ is the cosine similarity function, and τ is a temperature parameter. We use the max-pooling instead of average-pooling to aggregate the similarity scores, it focuses on the most relevant prompt in the library for each image, thereby capturing stronger visual-textual alignment (see Theorem 1 with the proof). The pre-trained source model serves as the student, its predictive distribution $p(y | x_i^t)$ is computed by applying the Softmax function to the final logits of the model.

Theorem 1 *For any concave function ϕ (such as the logarithm) and a random variable X , the following holds:*

$$\mathbb{E}[\phi(\mathbf{X})] \leq \phi(\mathbb{E}[\mathbf{X}]), \quad (2)$$

where $\mathbb{E}[\cdot]$ represents the mean of the random variable.

For each image x_i^t and class c_k , we generate a set of N text prompts $\{p_j^k\}_{j=1}^N$. This set of prompts can be viewed as a “bag” of evidence for class c_k . Each individual prompt p_j^k within the bag is an “instance”. The similarity score $s_j^k = S(I(x_i^t), T(p_j^k))$ represents the

alignment between the image and the j -th instance in the k -th bag. The goal is to aggregate the instance-level scores $\{s_j^k\}_{j=1}^N$ into a single bag-level score S_k for each class.

We introduce a latent variable $z \in \{1, \dots, N\}$ representing the index of the “best” explanatory text prompt. We are interested in the log marginal likelihood of the image belonging to class c_k , after marginalizing out z :

$$S_k = \log p(x_i^t \mid c_k) = \log \sum_{j=1}^N p(x \mid z = j, c_k) \cdot p(z = j). \quad (3)$$

Assuming a uniform prior $p(z = j) = 1/N$ and a likelihood related to the similarity score s_j^k , the objective becomes maximizing the Log-Sum-Exp (LSE) of similarity-related scores:

$$S_k = \log \sum_j \exp(L_j^k), \quad (4)$$

where $L_j^k \approx s_j^k$. Unfortunately, due to the stability and optimization efficiency issues associated with numerical overflow and non-sparse gradients, directly optimizing the LSE function is challenging. Instead, we can optimize a tractable lower bound, a core principle of maximizing the Evidence Lower Bound (ELBO).

Max-Pooling as a Tight Lower Bound. The LSE function is a smooth approximation of the maximum function, conversely, the maximum function serves as a tight, concave lower bound on the LSE objective:

$$S_k^{max} = \max_j \{L_j^k\} \leq S_k = \log \sum_j \exp(L_j^k). \quad (5)$$

Optimizing this tight lower bound is a valid and efficient strategy for optimizing the true objective S_k , especially when one prompt provides a significantly better explanation (i.e., one L_j^k dominates).

Average-Pooling as a Looser Lower Bound. To analyze the relationship between the average score and the LSE objective, we utilize the Theorem 1 to establish a bound. Let the random variable be e^L with a uniform distribution over the set $\{e^{L_j^k}\}_{j=1}^N$. Then:

$$\frac{1}{N} \sum_j L_j^k = \mathbb{E}[\log(e^L)] \leq \log(\mathbb{E}[e^L]) = \log\left(\frac{1}{N} \sum_j e^{L_j^k}\right), \quad (6)$$

rearranging this inequality yields:

$$\log \sum_j e^{L_j^k} \geq \log(N) + \frac{1}{N} \sum_j L_j^k. \quad (7)$$

This shows that the average log-score $S_k^{avg-log} = \frac{1}{N} \sum_j L_j^k$ is a significantly looser lower

bound on the true objective S_k , with a gap of at least $\log(N)$. Intuitively, averaging dilutes the signal from the best-matching prompt with noise from irrelevant ones.

3.3.2 Distillation Loss with Dynamic Weighting

We use the Kullback–Leibler (KL) divergence to align the predictive distributions between the teacher and student models. Importantly, not all predictions from the teacher model are equally reliable. To address this, we introduce an entropy-aware dynamic weighting mechanism that assigns higher weights to more confident (i.e., lower-entropy) predictions from CLIP. The dynamic weight α for a sample x_i^t is computed as follows:

$$\alpha(x_i^t) = \alpha_{\max} - (\alpha_{\max} - \alpha_{\min}) \cdot \frac{H(q(y | x_i^t))}{\log(K)}, \quad (8)$$

where $H(\cdot)$ indicates the entropy of the predicted distribution of the teacher, $\alpha_{\min}$ and $\alpha_{\max}$ define the allowable range of the dynamic weight. To further encourage the student model to produce confident predictions on the target domain, we augment the KL divergence-based distillation loss with an entropy minimization term. The cold-start distillation loss is defined as:

$$\begin{aligned} \mathcal{L}_{c-distill} = & \mathbb{E}_{x_i^t \sim D_T} [\alpha(x_i^t) \cdot D_{KL}(q(y | x_i^t) \| p(y | x_i^t))] \\ & + \lambda \cdot \mathbb{E}_{x_i^t \sim D_T} [H(p(y | x_i^t))], \end{aligned} \quad (9)$$

where $D_{KL}(q \| p) = \sum_{k=1}^K q_k \log \left(\frac{q_k}{p_k} \right)$. It represents the relative entropy of the probability distribution q with respect to the probability distribution p , measuring the difference in information between the two.

3.4 Pseudo-Source Domain Generation

3.4.1 Construction of the Pseudo-Source Domain Samples Index Set

From the previous distillation step, we obtain a more “Rational” target model θ_t . According to previous studies (Gou et al., 2021; Xu et al., 2020), different samples have varying instructional value, leading to different distillation outcomes. If we immediately assign pseudo-source domain samples based on the predictions of the target model-without considering the alignment or compatibility between the student and teacher models-this approach severs the intrinsic link between the quality of pseudo-source samples and the effectiveness of knowledge distillation. So, we propose a dual-model verification strategy to further enhance the quality of the pseudo-source samples. Specifically, we first compute the consistency score between the target model θ_t and the CLIP model θ_v , defined as the proportion of samples for which the two models predict the same label:

$$S_{consist} = \frac{1}{N_T} \sum_{i=1}^{N_T} \mathbb{I}(\hat{y}_M(x_i^t) = \hat{y}_{CLIP}(x_i^t)), \quad (10)$$

where i denotes the sample index, N_T is the total number of samples in the target domain, $\hat{y}_M$ is the label predicted by the target model, $\hat{y}_{CLIP}$ is the label predicted by the CLIP model, and $\mathbb{I}(\cdot)$ is the indicator function. We define a consistency threshold γ , and two sets: $A = \{i \mid \hat{y}_M(\mathbf{x}_i^t) = \hat{y}_{CLIP}(\mathbf{x}_i^t), \mathbf{x}_i^t \in D_T\}$ and $E_p = \{i \mid H_M(\mathbf{x}_i^t) < \eta_p, \mathbf{x}_i^t \in D_T\}$, where $\eta_p = \text{Percentile}(\{H_M(\mathbf{x}_j^t)\}_{j=1}^{N_T}, 100 - p)$, $H_M(\cdot)$ represents the prediction entropy produced by the target model θ_t , when the consistency score is high, we validate the effectiveness of the first stage of knowledge distillation and consider the entropy-based thresholding strategy to play a more auxiliary, safeguarding role. Accordingly, we relax the criteria for including samples in the set E_p and heuristically set the value of p to 25. Conversely, when the consistency score is low, we rely more heavily on the entropy-based strategy, and the new set $E_{p'}$ is restricted to only the top five percent of samples with the lowest entropy. Our final selection strategy for the pseudo-source domain samples index set I_{PS} is as follows:

$$I_{PS} = \begin{cases} A \cap E_p, & \text{if } S_{consist} \geq \gamma \\ E_{p'} \cup (A \cap E_p), & \text{if } S_{consist} < \gamma \end{cases} \quad (11)$$

where $E_{p'}$ represents a low-entropy set with a stricter threshold ($p' > p$). The value of γ will be explored in the analytical experiments section.

3.4.2 Class Balancing Strategy

Through the above steps, we obtain an initial set of high-quality pseudo-source domain samples. Prior research (Gulrajani & Lopez-Paz, 2020; Kumar et al., 2018) suggests that larger source domains generally lead to better transfer performance. However, our observations indicate that when the consistency score is low and the acceptance threshold for pseudo-source samples is raised accordingly, it may lead to class imbalance—some classes may contain very few or even no samples in the pseudo-source domain. This negatively impacts performance and even reduces the overall number of pseudo-source samples. To mitigate this issue, we first count the number of samples per class within the current pseudo-source index set I_{PS} . We then identify the underrepresented classes and, while ensuring that the total number of pseudo-source samples remains sufficiently large, we supplement these classes by selecting additional samples from the candidate pool based on entropy ranking—specifically, we choose the lowest-entropy samples for each deficient class. This process results in an updated pseudo-source index set I'_{PS} . We define the final pseudo-source domain and the remaining target domain as follows:

$$\begin{aligned} \hat{D}_S &= \{(x_i^t, \hat{y}_i) \mid i \in I'_{PS}\}, \\ \hat{D}_T &= \{(x_j^t) \mid j \in n^t \setminus I'_{PS}\}, \end{aligned} \quad (12)$$

where $\|I'_{PS}\| \geq 0.25\|n^t \setminus I'_{PS}\|$. It means that the final pseudo-source domain must contain no fewer than 20% of the total target domain samples. It is important to note that this threshold is not a finely tuned hyperparameter, but rather a conservative lower bound designed to ensure algorithmic robustness. It guarantees that even in extreme cases, the pseudo-source

domain retains a meaningful scale. In Stage 3, the aforementioned pseudo-source domain construction process is periodically revisited.

3.5 Model Adaptation

After constructing a robust pseudo-source domain $\hat{D}_S$ and retaining the remaining unlabeled target domain $\hat{D}_T$, we reformulate the original SFDA task as a pseudo-UDA problem. To address this challenge, in this section, we propose an adaptive training framework comprising three key components: robust pseudo-labels supervision, dual-policy domain alignment and consistency regularization.

3.5.1 Robust Pseudo-Labels Supervision

To enable the model to accurately classify samples from both the pseudo-source domain and the remaining target domain, we fine-tune the classifier using pseudo-labels. The classification loss is computed as follows:

$$\mathcal{L}_{\text{cls}} = \mathbb{E}_{(x_i^t, \hat{y}_i) \sim \hat{D}_S} [\mathcal{L}_{CE}(x_i^t, \hat{y}_i)] + \mathbb{E}_{x_j^t \sim \hat{D}_T} [\mathcal{L}_{CE}(x_j^t, \hat{y}_j)], \quad (13)$$

where $\mathcal{L}_{CE}$ denotes the standard cross-entropy loss. As mentioned above, $\hat{y}_i$ represents the pseudo-label for a pseudo-source domain sample, predicted by the target model θ_t . Since the remaining target domain samples are typically high-entropy for the target model, we generate their pseudo-labels $\hat{y}_j$ through a fusion of predictions from the target model and the CLIP model. The fused pseudo-labels are computed as follows:

$$\hat{y}_j = \arg \max_k [\omega \cdot p(y = k | x_j^t) + (1 - \omega) \cdot q(y = k | x_j^f)], \quad (14)$$

where ω represents the confidence weight of the prediction from the target model. The value of ω will be explored in the ablation experiment section.

3.5.2 Dual-Policy Domain Alignment Module

To reduce the distribution gap between the pseudo-source domain $\hat{D}_S$ and the remaining target domain $\hat{D}_T$, we introduce a complementary alignment strategy that combines implicit and explicit mechanisms. Specifically, we first incorporate a domain discriminator D to distinguish between features from the pseudo-source and target domains. Through adversarial training, the feature extractor G is encouraged to generate domain-invariant representations by making the features indistinguishable to the domain discriminator, thereby implicitly aligning local non-linear features. We then further integrate MMD (Maximum Mean Discrepancy) to explicitly minimize the feature distribution discrepancy between $\hat{D}_S$ and $\hat{D}_T$. The loss of our dual-policy domain alignment module is formulated as follows:

$$\begin{aligned} \min_{\theta_G} \max_{\theta_D} \mathcal{L}_{\text{adv}} = & \mathbb{E}_{x_i^t \sim \hat{D}_S} [\log D(G(x_i^t))] \\ & + \mathbb{E}_{x_j^t \sim \hat{D}_T} [\log(1 - D(G(x_j^t)))] + \mathcal{L}_{\text{MMD}}, \end{aligned} \quad (15)$$

where θ_G and θ_D are the parameters of the feature extractor and the domain discriminator. $\mathcal{L}_{MMD}$ is MMD feature alignment loss of the following form:

$$\mathcal{L}_{MMD} = \left\| \mathbb{E}_{x_i^t \sim \mathcal{D}_S} [\phi(G(x_i^t))] - \mathbb{E}_{x_j^t \sim \mathcal{D}_T} [\phi(G(x_j^t))] \right\|_{\mathcal{H}_k}^2, \quad (16)$$

where the function ϕ maps features into a reproducing kernel Hilbert space (RKHS), in which a Gaussian kernel is employed to measure the distribution discrepancy.

Input: Pre-trained source model θ_s ; unlabeled target domain D_T ; frozen CLIP θ_v ; distillation epochs $E_{distill}$; adaptation epochs E_{adapt} ; refresh interval R .

Output: The adapted target model θ'_t .

```

1: // === Stage 1: Cold-start Knowledge Distillation (run once) ===
2: Set  $\theta_t = \theta_s$ ;
3: for  $epoch = 1$  to  $E_{distill}$  do
4:   for each batch  $x_i^t$  from  $D_T$  do
5:     Obtain student predictions  $p(y|x_i^t)$  from  $\theta_t$ ;
6:     Obtain frozen-CLIP teacher predictions  $q(y|x_i^t)$  by Eq. (1);
7:     Compute dynamic weight  $\alpha$  by Eq. (8);
8:     Update  $\theta_t$  by minimizing  $\mathcal{L}_{c-distill}$  by Eq. (9);
9:   end for
10: end for
11: // === Stage 2: Pseudo-source Domain Generation ===
12: Infer on all  $x_i^t \in D_T$  with  $\theta_t$  and the frozen CLIP  $\theta_v$ ;
13: Generate  $\hat{D}_S$  and  $\hat{D}_T$ ;
14: // === Stage 3: Model Adaptation (Stages 2–3 alternated) ===
15: Set  $\theta'_t = \theta_t$ ,  $F'_T = F_T$ , init discriminator  $D$  and learnable prompt context  $\mathbf{V}$ ;
16: Fine-tuning  $\mathbf{V}$  on all  $D_T$  by minimizing  $\mathcal{L}_{mi}$  (Eq. (19)); obtain  $q'(y|x)$  (Eq. (18));
17: for  $epoch = 1$  to  $E_{adapt}$  do
18:   for each batch sampled from  $D_T$  do
19:     Partition batch into  $x_i^t \in \hat{D}_S$  and  $x_j^t \in \hat{D}_T$ ;
20:     Compute classification loss  $\mathcal{L}_{cls}$  by Eq. (13);
21:     Compute alignment loss  $\mathcal{L}_{adv}$  by Eq. (15);
22:     Compute regularization loss  $\mathcal{L}_{reg}$  by Eq. (17);
23:     Compute hard sample distillation  $\mathcal{L}_{h-distill}$  by Eq. (20) on  $\hat{D}_T$  from  $q'$ ;
24:     Update  $\theta'_t$  and  $D$  by minimizing  $\mathcal{L}_{total}$  by Eq. (22);
25:   end for
26:   if  $epoch \bmod R = 0$  and  $epoch < E_{adapt}$  then
27:     Fine-tuning  $\mathbf{V}$  on  $D_T$  (Eq. (19)) to refresh  $q'$  using the improved  $\theta'_t$ ;
28:     Regenerate  $\hat{D}_S, \hat{D}_T$  via the frozen CLIP  $\theta_v$  (Stage 2, Eqs. (10)–(12));
29:   end if
30: end for
31: return Adapted model  $\theta'_t$ .
```

Algorithm 1 Training of SRRI

3.5.3 Consistency Regularization

Directly updating the classifier F_T of the target model would clearly disrupt its learned knowledge structure, potentially degrading the quality of pseudo-labels and leading to instability during training. To mitigate this, following prior work, we introduce a new classifier F'_T , initialized from F_T , while keeping F_T frozen throughout training. We enforce consistency between the predictions of the current classifier F'_T and the frozen historical classifier F_T , promoting smoothness and stability in the training process:

$$\mathcal{L}_{\text{reg}} = \mathbb{E}_{x_i^t \sim D_T} [D_{JS}(p(y | x_i^t) \| p_{\text{old}}(y | x_i^t))], \quad (17)$$

where D_{JS} denotes the Jensen-Shannon divergence, and p_{old} represents the predictions produced by the historical classifier F_T .

3.5.4 Dynamic Pseudo-Source Domain Optimization

The cold-start distillation in Stage 1 relies on a generic frozen CLIP whose textual prompts are fixed. While this is sufficient to bootstrap the target model out of its biased starting point, these generic prompts are not well suited to the specific target task and therefore provide limited guidance on hard samples—i.e., the remaining target domain $\hat{D}_T$, which is precisely the subset the target model is still uncertain about. Therefore, following prompt tuning (Zhou et al., 2022), we replace the predefined templates of the teacher with a learnable context $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_M]$ prepended to each class name. The prompt for class c_k is $\mathbf{t}_k = [\mathbf{V}, \mathbf{e}_k]$, where $\mathbf{e}_k$ is the token embedding of the k -th class name, and the CLIP distribution is:

$$q'(y = c_k | x_i^t) = \text{Softmax}(\tau \cdot S(I(x_i^t), T(\mathbf{t}_k))), \quad (18)$$

in which, only $\mathbf{V}$ is trainable. since the image features are produced by a frozen encoder, they are computed once and cached, and each optimization step only re-encodes the K class prompts, making fine-tune inexpensive.

As the target labels are unavailable, we optimize $\mathbf{V}$ on all target samples by maximizing the mutual information between the target-model predictions $p(y | x_i^t)$ and the CLIP predictions $q'(y | x_i^t)$, encouraging the two views to agree. Estimating the joint distribution over a batch $\mathcal{B}$ as $P = \frac{1}{|\mathcal{B}|} \sum p(y | x_i^t), q'(y | x_i^t)^\top$ and symmetrizing it as $\tilde{P} = \frac{1}{2}(P + P^\top)$, meanwhile, to prevent prompt learning from collapsing to a few dominant classes, we constrain the average predictions $\bar{q}'$ to follow a uniform distribution $\mathcal{U}$. The mutual-information loss is:

$$\mathcal{L}_{mi} = - \sum_{a=1}^K \sum_{b=1}^K \tilde{P}_{ab} \log \frac{\tilde{P}_{ab}}{\tilde{P}_a \cdot \tilde{P}_b} + \beta D_{KL}(\bar{q}' \| \mathcal{U}), \quad (19)$$

where $\tilde{P}_a$ and $\tilde{P}_b$ are the marginals of $\tilde{P}$, β serves as a trade-off parameter. Then, we distill its knowledge into the target model on the hard samples $\hat{D}_T$. The hard sample distillation loss is:

$$\mathcal{L}_{h\text{-}distill} = \mathbb{E}_{x_j^t \sim \hat{D}_T} \left[\alpha(x_j^t) \cdot D_{KL}(q'(y | x_j^t) \| p(y | x_j^t)) \right]. \quad (20)$$

As the target model improves over training, the relative value of the teacher diminishes, we therefore anneal the weight of this term linearly, $\lambda_{hd}(t) = 1 - t/T$, where t and T denote the current and total adaptation iterations. In summary, distillation loss is defined as follows:

$$\mathcal{L}_{distill} = \begin{cases} \mathcal{L}_{c\text{-}distill}, & \text{Stage 1 (cold-start, executed once)} \\ \lambda_{hd}(t) \mathcal{L}_{h\text{-}distill}, & \text{Stage 3 (adaptation, } t \geq 1) \end{cases} \quad (21)$$

Combining the losses described above, the overall objective function for adapting the target model is formulated as:

$$\mathcal{L}_{total} = \mathcal{L}_{distill} + \mathcal{L}_{cls} + \mathcal{L}_{reg} + \lambda_a \mathcal{L}_{adv}, \quad (22)$$

where $\lambda_a > 0$ is a trade-off parameter. In our training pipeline, $\mathcal{L}_{c\text{-}distill}$ is executed only once. Subsequently, $\mathcal{L}_{h\text{-}distill}$ is periodically re-executed on the updated $\hat{D}_T$ at fixed epoch interval R , after which the pseudo-source domain is reconstructed. We summarize the whole training procedure of SRRI in Algorithm 1.

4 Experiments

In this section, we first introduce the datasets and experimental setup. Then, we take the state-of-the-art methods that are currently available in the field of SFDA that can be used as comparators to perform comparative experiments on 4 domain adaptation benchmark datasets. And we purposely compare the current representative pseudo-source domain based SFDA method to show the superiority of our SRRI method. In addition, ablation experiments, feature visualization, Grad-CAM (Selvaraju et al., 2017) visualization, parametric analysis, and pseudo-source domain quality analysis are performed to further validate the effectiveness of each element of the proposed methodology as well as the elimination of spurious correlations.

4.1 Experimental Setup

4.1.1 Datasets

We evaluate SRRI on four widely used domain adaptation benchmark datasets. (1) **Office-31** (Saenko et al., 2010) is a small-scale dataset comprising three distinct domains and 31 categories of office-related objects. (2) **Office-Home** (Venkateswara et al., 2017) is a widely recognized medium-scale dataset that includes four diverse domains and 65 object categories. (3) **VisDA** (Peng et al., 2017) is a challenging benchmark with 12 categories, where the source domain consists of synthetic images and the target domain comprises real images. (4) **DomainNet-126** (Peng et al., 2019) is a large-scale dataset covering four visually diverse domains and a total of 126 categories.

4.1.2 Competitor

We compare SRRI with existing high-performing methods, categorized into four groups. (1) The first group includes the predictions from the source model without any adaptation and the CLIP Zero-shot (ViT-B/32) (Radford et al., 2021). (2) The second group consists of six representative UDA methods: BNM (Cui et al., 2020), CDAN (Long et al., 2018), SFAN (Xu et al., 2019), BCDM (Li et al., 2021), MCD (Saito et al., 2018), and DAPL (Ge et al., 2023). (3) The third group comprises 18 state-of-the-art SFDA methods, including SHOT (Liang et al., 2020), NRC (Yang et al., 2021), GKD (Tang et al., 2021), PS (Du et al., 2024), U-SFAN+ (Roy et al., 2022), AdaCon (Chen et al., 2022), CoWA (Lee et al., 2022), SCLM (Tang et al., 2022), AAA (Li et al., 2022a), VDM (Tian et al., 2022), DIPE (Wang et al., 2022), PLUE (Litrico et al., 2023), GAP (Chhabra et al., 2023), SFADA (He et al., 2024), ASOGE (Cui et al., 2023), DPLS (Ma et al., 2024), CPD (Zhou et al., 2024), and TPDS (Tang et al., 2024a). (4) The fourth group comprises 4 CLIP-based SFDA methods, including DIFO (Tang et al., 2024b), ViLAaD++ (Tarashima et al., 2025), SHOT w/ Co-learn++ (Zhang et al., 2025), and NRC w/Co-learn++ (Zhang et al., 2025). DIFO and ViLAaD++ employ the ViT-B/32 variant of CLIP, whereas SHOT w/Co-learn++ and NRC w/Co-learn++ adopt the ViT-L/14@336 variant. Notably, ViT-L/14@336 exhibits superior baseline generalization performance compared with ViT-B/32. Among them, PS is the current representative SFDA method based on pseudo-source domain.

4.1.3 Implementation Details

For all experiments, we conduct in PyTorch using a single NVIDIA GeForce RTX 4090 GPU. We first pretrain the model on the labeled source domain to simulate the “source model” provided by the user in a typical SFDA scenario. The model checkpoint that achieves the best performance on the source-domain validation set is saved as the starting point for subsequent adaptation. We adopt ResNet-50 or ResNet-101 as the backbone feature extractor, followed by a bottleneck layer and a classifier. The bottleneck layer is set to have a dimensionality of 256. For the teacher model in our approach, we utilize the ViT-B/32 variant of CLIP, with its parameters kept frozen throughout all stages. All experiments are conducted using the PyTorch framework. Optimization is performed using stochastic gradient descent (SGD) with Nesterov momentum. The momentum is set to 0.9, and the weight decay is $5e-4$. The learning rates for the domain discriminator and classifier are set to be 10 times that of the feature extractor. Specifically, the learning rate is $5e-4$ for DomainNet-126 and $1e-3$ for all other datasets. The number of training epochs is set to 200 for Office-31, 100 for Office-Home, 15 for VisDA, and 40 for DomainNet-126. For the cold-start distillation stage, the number of epochs is set to 20 for Office-31 and Office-Home, 3 for VisDA, and 5 for DomainNet-126. For the fixed epoch interval, it is set to 20 for Office-31 and Office-Home, 3 for VisDA, and 8 for DomainNet-126. Regarding hyperparameters, the adversarial loss weight λ_{adv} is set to 0.1, and the batch size is fixed at 128 for all datasets.

4.2 Experimental Results

The comparison results across the four evaluation datasets are presented in Tables 1, 2, 3, 4. In terms of average accuracy, SRRI outperforms the previous best-performing meth-

Table 1 Classification accuracy (%) on Office-Home (ResNet-50)

Method	SF	ViL	A→C	A→P	A→R	C→A	C→P	C→R	P→A	P→C	P→R	R→A	R→C	R→P	Avg.
Source	-	-	43.7	67.0	73.9	49.9	60.1	62.5	51.7	40.9	72.6	64.2	46.3	78.1	59.2
CLIP Zero-shot (ICML'21)	-	✓	60.2	84.5	86.0	75.1	84.5	85.6	74.4	60.1	85.0	74.5	59.8	83.7	76.1
CDAN (NIPS'18)	✓	✓	50.7	70.6	76.0	57.6	70.0	70.0	57.4	50.9	77.3	70.9	56.7	81.6	65.8
SEAN (ICCV'19)	✓	✓	52.0	71.7	76.3	64.2	69.9	71.9	63.7	51.4	77.1	70.9	57.1	81.5	67.3
BNM (CVPR'20)	✓	✓	52.3	73.9	80.0	63.3	72.9	74.9	61.7	49.5	79.7	70.5	53.6	82.2	67.9
SHOT (ICML'20)	✓	✓	57.1	78.1	81.5	68.0	78.2	78.1	67.4	54.9	82.2	73.3	58.8	84.3	71.8
NRC (NIPS'21)	✓	✓	57.7	80.3	82.0	68.1	79.8	78.6	65.3	56.4	83.0	71.0	58.6	85.6	72.2
GKD (IJROS'21)	✓	✓	56.5	78.2	81.8	68.7	78.9	79.1	67.6	54.8	82.6	74.4	58.5	84.8	72.2
PS (ML'21)	✓	✓	57.8	77.3	81.2	68.4	76.9	78.1	67.8	57.3	82.1	75.2	59.1	83.4	72.1
U-SFAN+ (CV'22)	✓	✓	57.8	77.8	81.6	67.9	77.3	79.2	67.2	54.7	81.2	73.3	60.3	83.9	71.9
AdaCon (CVPR'22)	✓	✓	47.2	75.1	75.5	60.7	73.3	73.2	60.2	45.2	76.6	65.6	48.3	79.1	65.0
AAA (TPAMI'22)	✓	✓	56.7	78.3	82.1	66.4	78.5	79.4	67.6	53.5	81.6	74.5	58.4	84.1	71.8
VDM (TCSVT'22)	✓	✓	59.3	75.3	78.3	67.6	76.0	75.9	68.8	57.7	79.6	74.0	61.1	83.6	71.4
DIPE (CVPR'22)	✓	✓	56.5	79.2	80.7	70.1	79.8	78.8	67.9	55.1	83.5	74.1	59.3	84.8	72.5
PLUE (CVPR'23)	✓	✓	49.1	73.5	78.2	62.9	73.5	74.5	62.2	48.3	78.6	68.6	51.8	81.5	66.9
GAP (WACV'23)	✓	✓	55.4	73.4	80.8	67.2	75.5	78.3	65.5	54.0	82.4	74.3	59.4	84.0	70.8
SFADA (PR'24)	✓	✓	56.1	78.0	81.6	68.5	79.5	78.5	67.8	56.0	82.3	73.6	57.8	83.0	71.9
ASOGE (TCSVT'24)	✓	✓	59.1	78.4	81.0	67.7	78.4	77.5	65.8	57.2	80.2	72.7	60.7	83.3	71.8
DPLS (PR'24)	✓	✓	57.8	79.0	81.6	67.8	79.9	79.1	68.6	54.7	82.5	72.5	59.6	85.9	72.4
DIFO (CVPR'24)	✓	✓	70.6	90.6	88.8	82.5	90.6	88.8	80.9	70.1	88.9	83.4	70.5	91.2	83.1
ViLAaD++ (arXiv'25)	✓	✓	70.1	91.6	89.9	83.2	92.0	90.0	81.0	71.7	89.9	83.3	71.3	92.3	83.9
SHOT w/Co-learn++ (IJCV'25)	✓	✓	62.2	83.1	84.9	71.5	81.7	81.7	70.9	61.9	84.1	75.9	65.5	86.6	75.8
NRC w/Co-learn++ (IJCV'25)	✓	✓	76.4	88.8	88.6	82.4	89.1	88.2	81.0	75.2	89.7	82.4	76.3	91.9	84.2
SRRI (ours)	✓	✓	72.9±	90.5±	91.1±	83.8±	91.1±	90.9±	82.8±	73.9±	91.3±	82.8±	73.7±	90.9±	84.6±
			0.2	0.1	0.3	0.2	0.1	0.2	0.3	0.2	0.4	0.1	0.2	0.1	0.2

Bold text indicates the best results. SF means source-free, and ViL means whether using ViL model

Table 2 Classification accuracy (%) on VisDA (ResNet-101)

Method	SF	ViL	pla.	bic.	bus	car	hor.	kni.	mot.	per.	pla.	ska.	tra.	tru.	Avg.
Source	-	-	60.7	21.7	50.8	68.5	71.8	5.4	86.4	20.2	67.1	43.3	83.3	10.6	49.2
CLIP Zero-shot (ICML'21)	-	✓	93.2	87.1	82.6	69.0	90.4	88.2	87.7	76.1	75.4	89.3	87.5	69.9	83.0
CDAN (NIPS'18)	✓	✓	85.2	66.9	83.0	50.8	84.2	74.9	88.1	74.5	83.4	76.0	81.9	38.0	73.9
SEAN (ICCV'19)	✓	✓	93.6	61.3	84.1	70.6	94.1	79.0	91.8	79.6	89.9	55.6	89.0	24.4	76.1
BCDM (AAAI'21)	✓	✓	95.1	87.6	81.2	73.2	92.7	95.4	86.9	82.5	95.1	84.8	88.1	39.5	83.4
SHOT (ICML'20)	✓	✓	94.3	88.5	80.1	57.3	93.1	94.9	80.7	80.3	91.5	89.1	86.3	58.2	82.9
NRC (NIPS'21)	✓	✓	96.8	91.3	82.4	62.4	96.2	95.9	86.1	80.6	94.8	94.1	90.4	68.2	85.9
GKD (IJOS'21)	✓	✓	95.3	87.6	81.7	58.1	93.9	94.0	80.0	80.0	91.2	91.0	86.9	56.1	83.0
PS (ML'21)	✓	✓	95.2	86.2	82.1	61.5	93.2	95.6	86.7	80.4	91.5	90.8	85.8	59.3	84.0
U-SFAN+ (CV'22)	✓	✓	94.9	87.4	78.0	56.4	93.8	95.1	80.5	79.9	90.1	90.1	85.3	60.4	82.7
SCLM (NN'22)	✓	✓	97.1	90.7	85.6	62.0	97.3	94.6	81.8	84.3	93.6	92.8	88.0	55.9	85.3
AAA (TPAMI'22)	✓	✓	94.4	85.9	74.9	60.2	96.0	93.5	87.8	80.8	90.2	92.0	86.6	68.3	84.2
VDM (TCSVT'22)	✓	✓	96.9	90.0	80.0	64.4	96.8	96.4	86.7	83.3	96.2	87.9	89.8	54.7	85.3
DIPE (CVPR'22)	✓	✓	95.2	87.6	78.8	55.9	93.9	95.0	84.1	81.7	92.1	88.9	85.4	58.0	83.1
GAP (WACV'23)	✓	✓	94.2	84.8	82.5	57.2	93.8	95.1	86.5	78.2	83.1	87.8	86.3	53.5	81.9
SFADA (PR'24)	✓	✓	94.2	79.6	79.8	65.7	92.6	94.1	87.3	80.8	88.1	91.4	83.3	55.0	82.7
ASOGE (TCSVT'24)	✓	✓	94.9	84.3	76.8	54.3	94.9	93.4	86.0	85.0	87.2	90.0	86.7	62.7	83.2
CPD (PR'24)	✓	✓	96.7	88.5	79.6	69.0	95.9	96.3	87.3	83.3	94.4	92.9	87.0	58.7	85.8
DIFO (CVPR'24)	✓	✓	97.5	89.0	90.8	83.5	97.8	97.3	93.2	83.5	95.2	96.8	93.7	65.9	90.3
ViLAaD++ (arXiv'25)	✓	✓	98.1	90.3	88.2	79.2	97.5	97.5	92.9	84.8	92.0	96.8	94.0	75.3	90.5
SHOT w/Co-learn++ (IJCV'25)	✓	✓	97.2	91.3	83.8	69.1	97.1	98.0	88.9	83.0	91.5	94.6	89.3	57.6	86.8
NRC w/Co-learn++ (IJCV'25)	✓	✓	98.0	90.8	83.9	69.0	97.4	97.6	91.7	81.6	92.8	96.2	92.8	59.9	87.6
SRRI (ours)	✓	✓	98.9±	92.1±	90.6±	76.9±	98.3±	97.5±	94.5±	85.3±	91.9±	95.8±	95.2±	75.0±	91.0±
			0.1	0.2	0.3	0.3	0.1	0.2	0.3	0.2	0.2	0.1	0.2	0.3	0.2

Bold text indicates the best results. SF means source-free, and ViL means whether using ViL model

Table 3 Classification accuracy (%) on DomainNet-126 (ResNet-50)

Method	SF	ViL	C→P	C→R	C→S	P→C	P→R	P→S	R→C	R→P	R→S	S→C	S→P	S→R	Avg.
Source	-	-	44.6	59.8	47.5	53.3	75.3	46.2	55.3	62.7	46.4	55.1	50.7	59.5	54.7
CLIP Zero-shot (ICML'21)	-	✓	73.6	85.9	70.8	75.2	86.1	71.1	74.9	73.8	71.8	74.7	73.9	86.5	76.5
MCD (CVPR'18)	✗	✗	56.6	79.7	53.6	58.3	83.3	60.9	61.9	69.3	56.2	56.2	66.7	81.7	65.4
DAPL (TNNLS'23)	✗	✗	72.4	87.6	65.9	72.7	87.6	65.6	73.2	72.4	66.2	73.8	72.9	87.8	74.8
SHOT (ICML'20)	✓	✗	63.5	78.2	59.5	67.9	81.3	61.7	67.7	67.6	57.8	70.2	64.0	78.0	68.1
NRC (NIPS'21)	✓	✗	62.6	77.1	58.3	62.9	81.3	60.7	64.7	69.4	58.7	69.4	65.8	78.7	67.5
GKD (IROS'21)	✓	✗	61.4	77.4	60.3	69.6	81.4	63.2	68.3	68.4	59.5	71.5	65.2	77.6	68.7
AdaCon (CVPR'22)	✓	✗	60.8	74.8	55.9	62.2	78.3	58.2	63.1	68.1	55.6	67.1	66.0	75.4	65.4
CoWA (ICML'22)	✓	✗	64.6	80.6	60.6	66.2	79.8	60.8	69.0	67.2	60.0	69.0	65.8	79.9	68.6
PLUE (CVPR'23)	✓	✗	59.8	74.0	56.0	61.6	78.5	57.9	61.6	65.9	53.8	67.5	64.3	76.0	64.7
TPDS (IJCV'24)	✓	✗	62.9	77.1	59.8	65.6	79.0	61.5	66.4	67.0	58.2	68.6	64.3	75.3	67.1
DIFO (CVPR'24)	✓	✓	76.6	87.2	74.9	80.0	87.4	75.6	80.8	77.3	75.5	80.5	76.7	87.3	80.0
ViLAaD++ (arXiv'25)	✓	✓	80.3	91.4	77.7	83.8	91.5	78.2	84.1	80.8	77.6	84.2	80.2	91.3	83.4
SRRI (ours)	✓	✓	81.6	91.9	79.7	84.5	91.9	79.7	84.7	81.8	79.6	85.2	81.7	91.8	84.5

Bold text indicates the best results. SF means source-free, and ViL means whether using ViL model

Table 4 Classification accuracy (%) on Office-31 (ResNet-50)

Method	SF	ViL	A→D	A→W	D→A	D→W	W→A	W→D	Avg.
Source	-	-	79.1	76.6	59.9	95.5	61.4	98.8	78.6
CLIP-ZS	-	✓	82.5	80.3	76.0	80.7	76.5	83.6	79.9
CDAN	✗	✗	92.9	94.1	71.0	98.6	69.3	100.	87.7
SFAN	✗	✗	90.7	90.1	73.0	98.6	70.2	99.8	87.1
BNM	✗	✗	90.3	91.5	98.5	100.	70.9	71.6	87.1
BCDM	✗	✗	95.4	93.8	73.1	98.6	73.0	100.	89.0
SHOT	✓	✗	93.7	91.1	74.2	98.2	74.6	100.	88.6
NRC	✓	✗	96.0	90.8	75.3	99.0	75.0	100.	89.4
GKD	✓	✗	94.6	91.6	75.1	98.7	75.1	100.	89.2
U-SFAN+	✓	✗	94.2	92.8	74.6	98.0	74.4	99.0	88.8
AdaCon	✓	✗	87.7	83.1	73.7	91.3	77.6	72.8	81.0
SCLM	✓	✗	95.8	90.0	75.5	98.9	75.5	99.8	89.4
PLUE	✓	✗	89.2	88.4	72.8	97.1	69.6	97.9	85.8
GAP	✓	✗	90.6	90.9	74.5	98.7	73.9	99.8	88.1
SFADA	✓	✗	94.8	92.0	76.5	97.6	75.7	99.8	89.4
ASOGE	✓	✗	95.6	94.1	74.3	98.1	74.2	99.7	89.3
DIFO	✓	✓	97.2	95.5	83.0	97.2	83.2	98.8	92.5
ViLAaD++	✓	✓	96.6	95.1	85.1	96.0	83.7	99.0	92.6
SHOT w/Co-learn++	✓	✓	96.2	94.7	78.3	98.2	77.6	99.8	90.8
NRC w/Co-learn++	✓	✓	96.4	95.8	78.7	98.9	78.5	99.8	91.4
SRRI (ours)	✓	✓	97.1±	94.9±	83.9±	99.1±	83.3±	100. ±	93.1 ±
			0.2	0.3	0.1	0.2	0.2	0.0	0.2

Bold text indicates the best results. SF means source-free, and ViL means whether using ViL model

ods on each dataset, surpassing ViLAaD++ on Office-31 by 0.5%, NRC w/Co-learn++ on Office-Home by 0.4%, ViLAaD++ on VisDA by 0.5%, and ViLAaD++ on DomainNet-126 by a significant margin of 1.1%. Specifically, SRRI achieves the best performance on the W2D task in Office-31, ranks first in five out of twelve tasks in Office-Home, secures the top results in six tasks on VisDA, and achieves the best performance in twelve tasks on the more challenging DomainNet-126 benchmark. Notably, compared to the PS method, SRRI achieves superior performance on both Office-Home and VisDA, indicating that SRRI generates a higher-quality pseudo-source domain. Besides, we evaluate SRRI using three different random seeds on Office31, OfficeHome and VisDA, and report the mean and standard deviation across the three runs. As shown, SRRI exhibits consistently low standard deviations on all three datasets, demonstrating its robustness to different random seed initializations. Collectively, these results strongly demonstrate the effectiveness of SRRI in the SFDA setting.

4.3 Analytical Experiments

4.3.1 Ablation Study

We present the ablation study of SRRI on the Office-31, Office-Home, and VisDA datasets in Table 5. First, we analyze the main objectives. Without any adaptation, the source model achieves only 62.3% average accuracy. Using $\mathcal{L}_{distill}$ alone significantly improves

Table 5 Classification accuracy (%) of ablation study on Office-31, Office-Home and VisDA

$\mathcal{L}_{adv}$	$\mathcal{L}_{cls}$	$\mathcal{L}_{reg}$	$\mathcal{L}_{distill}$	Office-31	Office-Home	VisDA	Avg.
$\times$	$\times$	$\times$	$\times$	78.6	59.2	49.2	62.3
$\times$	$\times$	$\times$	$\checkmark$	85.7	79.2	86.9	83.9
$\checkmark$	$\times$	$\times$	$\times$	81.8	66.2	79.8	75.9
$\times$	$\checkmark$	$\times$	$\times$	82.4	68.5	80.1	77.0
$\times$	$\checkmark$	$\checkmark$	$\times$	82.8	69.6	81.0	77.8
$\checkmark$	$\checkmark$	$\times$	$\times$	83.0	70.8	82.7	78.8
$\checkmark$	$\times$	$\checkmark$	$\times$	81.5	67.6	80.7	76.6
$\times$	$\checkmark$	$\times$	$\checkmark$	89.1	80.5	87.5	85.7
$\checkmark$	$\checkmark$	$\checkmark$	$\times$	84.3	71.3	84.2	79.9
$\times$	$\checkmark$	$\checkmark$	$\checkmark$	91.2	82.8	88.5	87.5
$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	93.1	84.6	91.0	89.6
SRRI w/o $\alpha(\cdot)$				92.5	83.9	90.1	88.8
SRRI w/o dual-model validation				91.6	82.3	88.2	87.4
SRRI w/o class balancing				92.0	82.9	89.0	88.0
$\mathcal{L}_{adv}$ w/o $\mathcal{L}_{MMD}$				92.4	83.5	89.7	88.5
SRRI w/o $\mathcal{L}_{c-distill}$				91.0	81.6	87.5	86.7
SRRI w/o $\mathcal{L}_{h-distill}$				91.2	82.4	87.9	87.2
SRRI w/o PS reconstruction				92.1	83.6	89.3	88.3
SRRI w/o DPDO				90.6	82.4	86.7	86.6
DPDO w/o prompt fine-tuning				91.5	83.0	88.2	87.6
SRRI w/o DPDO & $\mathcal{L}_{c-distill}$				87.5	72.0	85.8	81.8

$\mathcal{L}_{distill}$ includes both $\mathcal{L}_{c-distill}$ and $\mathcal{L}_{h-distill}$. DPDO denotes the proposed Dynamic Pseudo-source Domain Optimization

the average accuracy to 83.9%, showing that CLIP-based de-biasing provides an effective independent adaptation signal. In contrast, the combination of $\mathcal{L}_{adv}$, $\mathcal{L}_{cls}$, and $\mathcal{L}_{reg}$ without distillation obtains 79.9%, which is lower than $\mathcal{L}_{distill}$ alone. This indicates that the improvement of SRRI does not merely come from stacking standard adaptation losses, but relies strongly on the rational knowledge introduced by CLIP. When all objectives are combined, SRRI achieves the best average accuracy of 89.6%. Second, we further ablate key designs in SRRI. Removing the entropy-aware weight $\alpha(\cdot)$, dual-model validation, class balancing, and $\mathcal{L}_{MMD}$ reduces the average accuracy to 88.8%, 87.4%, 88.0%, and 88.5%, respectively, verifying the effectiveness of each component in pseudo-source construction and domain alignment. Removing $\mathcal{L}_{c-distill}$ and $\mathcal{L}_{h-distill}$ decreases the average accuracy to 86.7% and 87.2%, respectively, indicating that cold-start CLIP distillation plays a more fundamental role, while hard sample distillation further refines the target model during adaptation. In addition, removing pseudo-source reconstruction or the whole DPDO module reduces the performance to 88.3% and 86.6%, removing the prompt tuning component from DPDO, thereby relying on a fully frozen CLIP to guide hard sample distillation, reduces the performance to 87.6%. When both DPDO and $\mathcal{L}_{c-distill}$ are removed, the accuracy drops sharply to 81.8%, demonstrating that CLIP distillation, rather than CLIP verification alone, is crucial for SRRI. Overall, these results confirm that the performance gain comes from the cooperation between CLIP-guided de-biasing, reliable pseudo-source construction, and dynamic pseudo-source optimization.

4.3.2 Grad-CAM Visualization

We visualize the GradCAM of PS and SRRI trained on VisDA to show the ability of SRRI to eliminate spurious correlations in Fig. 3. PS is currently a representative pseudo-source domain based SFDA method. For PS, we utilize the pre-trained source model to obtain GradCAM for source domain samples and pseudo-source domain samples, respectively. For SRRI, we use the pretrained source model to obtain the Grad-CAM visualizations of the source-domain samples. After cold-start knowledge distillation, we then use the target model to generate the Grad-CAM visualizations of the pseudo-source domain samples from the first construction. The results are shown in the first two rows of the figure. We note that, on source domain samples, both are able to exploit their significant appearance differences (overall fuselage of the plane versus long straight knife) to perfectly predict both categories. However, PS seems to be influenced by this spurious correlation of the long and straight appearance of knives in the source domain sample, causing it to focus excessively on certain long and straight parts of the planes in the target domain sample instead of the overall fuselage, and thus incorrectly predicting them as knives. SRRI corrects this error by focusing instead on the overall fuselage, resulting in correct predictions. Furthermore, the third row of the figure demonstrates the effectiveness of the proposed dynamic pseudo-source domain optimization module on the challenging car and truck categories. Specifically, we treat the pseudo-source domain constructed immediately after cold-start distillation as the initial pseudo-source domain, and the one obtained after the final reconstruction as the final pseudo-source domain. Although spurious correlations lead to incorrect predictions on several hard samples in the initial pseudo-source domain, these errors are effectively corrected in the final pseudo-source domain through task-specific prompt tuning and subsequent knowledge distillation.

4.3.3 Hyperparameter Analysis

Figure 4 illustrates the accuracy under different values of the consistency threshold γ in Eq. (11) and the confidence threshold ω of predictions from the target model in Eq. (14). The results show that a low consistency threshold degrades performance, which can be attrib-

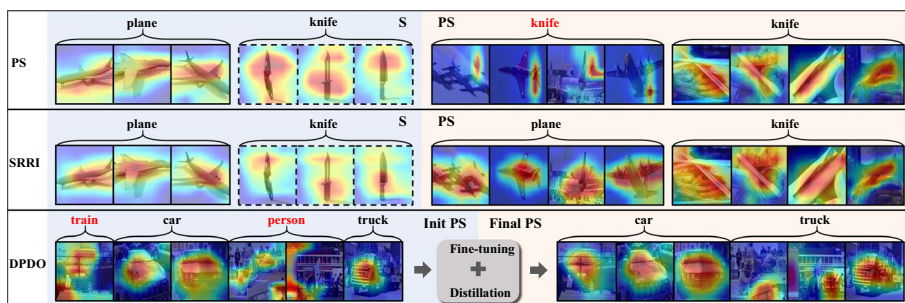


Fig. 3 GradCAM on VisDA using PS and the proposed SRRI. The plane and knife in the real label are represented by solid and dashed boxes, respectively. Predictions of the model are displayed at the top of each row. Red color indicates incorrect predictions. Regarding the first two rows, light blue and light red color indicate training samples and pseudo-source domain samples. Regarding the third row, DPDO means Dynamic Pseudo-source Domain Optimization, light blue and light red represent the first and last pseudo-source domain samples obtained, respectively

Fig. 4 Average performance on the Office-Home dataset with different combinations of γ and ω

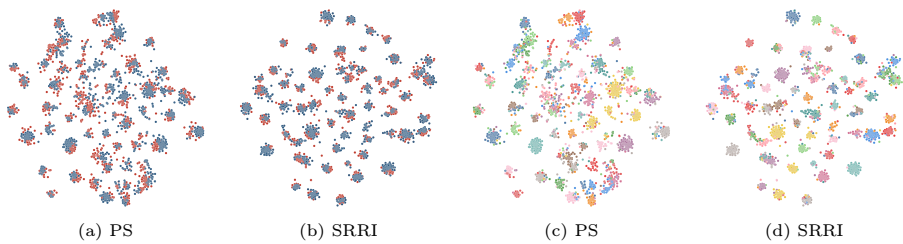
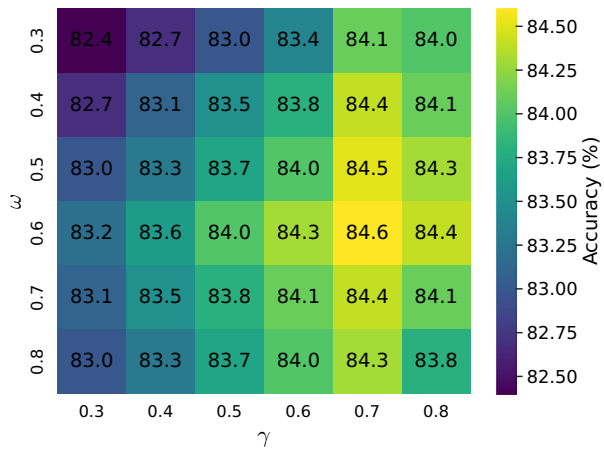


Fig. 5 Feature alignment and classification visualization comparison on transfer task P→C in Office-Home. **a, b**: feature alignment, blue is source feature and red is target feature. **c, d**: feature classification

uted to the lack of class balancing strategy when the consistency score between the student and teacher models is low—further validating the effectiveness of our pseudo-source domain generation strategy. In addition, both overly high and overly low ω values adversely affect the quality of pseudo-labels. Assigning higher confidence to the target model and using the CLIP model as guidance lead to more stable performance.

4.3.4 Feature Alignment and Classification Visualization

As illustrated in Fig. 5, we conduct a T-SNE feature visualization to compare the classic PS model with our proposed SRRI method. In terms of domain alignment, SRRI demonstrates a much more uniform and thorough mixing of source and target domain samples. In contrast, while some degree of mixing is observed in PS, large, distinct clusters of source (blue) and target (red) samples remain clearly visible. Regarding class separability, SRRI maintains strong intra-class compactness, with features from the same category forming well-defined clusters.

4.3.5 Pseudo-Source Domain Quality Analysis

To directly evaluate the quality of the constructed pseudo-source domain, we further report the pseudo-label accuracy of pseudo-source samples during adaptation. As shown in Fig.

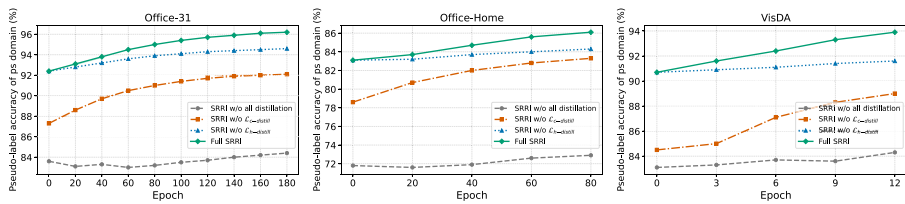


Fig. 6 Pseudo-label accuracy of the pseudo-source (ps) domain during adaptation on Office-31, Office-Home and VisDA

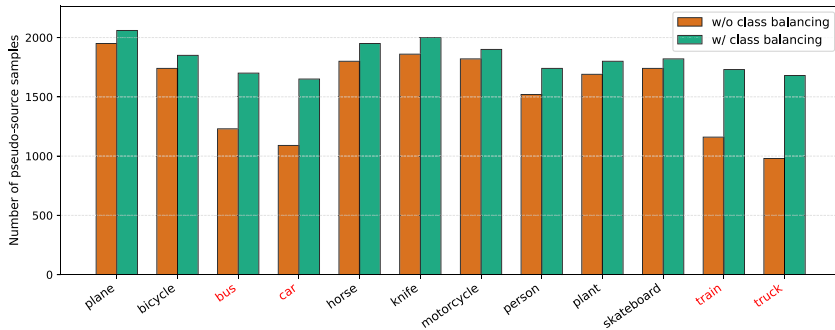


Fig. 7 Class distribution of the generated pseudo-source domain on VisDA. Red denotes hard class samples

6, Full SRRI consistently achieves the highest pseudo-source pseudo-label accuracy on Office-31, Office-Home, and VisDA. On Office-31, the pseudo-label accuracy of Full SRRI increases from 92.4% to 96.2%, while the variant without $\mathcal{L}_{h-distill}$ reaches only 94.6%. Similar trends can be observed on Office-Home and VisDA. These results indicate that the dynamic pseudo-source optimization process continuously improves the reliability of pseudo-source samples. When all distillation terms are removed, the pseudo-label accuracy remains much lower on all datasets, removing $\mathcal{L}_{c-distill}$ also causes a clear degradation, especially in the early stage. In contrast, removing $\mathcal{L}_{h-distill}$ mainly weakens the later improvement. We also analyze the class-wise distribution of pseudo-source samples on VisDA to evaluate the effect of the class balancing strategy. As shown in Fig. 7, without class balancing, several hard classes such as bus, car, train, and truck contain much fewer pseudo-source samples. After applying class balancing, the number of samples in these classes is substantially increased, e.g., truck increases from 980 to 1680, car from 1090 to 1650, train from 1160 to 1730, and bus from 1230 to 1700. Quantitatively, the max/min class-count ratio decreases from 1.99 to 1.25, and the standard deviation of class-wise sample counts decreases from 326.0 to 126.4. These results show that the proposed class balancing strategy effectively improves the coverage and balance of the pseudo-source domain.

5 Conclusion

In this paper, we analyzed a critical issue in existing SFDA methods that constructed pseudo-source domains: the confirmation bias of the source model, induced by spurious inter-domain correlations, which compromised the reliability of the generated pseudo-source domain. We proposed a novel Staying Rational and Resisting Illusions (SRRI) method to rectify the misclassification tendencies of the source model on pseudo-source samples by distilling the powerful prior knowledge from the CLIP model. Next, we designed a pseudo-source domain generation strategy that integrated CLIP-guided dual-model validation and class balancing strategy. This ensured the creation of a superior pseudo-source domain even in scenarios with sub-optimal distillation performance. Furthermore, we proposed a dynamic pseudo-source domain optimization mechanism that continuously fine-tuned task-specific prompts of the CLIP during adaptation to provide an updated teacher signal for guiding the target model on hard samples, thereby periodically reconstructing the pseudo-source domain. Extensive experiments on four datasets demonstrate that our method not only significantly outperforms existing state-of-the-art approaches but also generates pseudo-source domains of substantially higher-quality. We believe our work offers an effective and promising solution to the challenges posed by the unavailability of source data in SFDA. For future work, we plan to extend this framework to multi-source free domain adaptation scenarios and explore more lightweight distillation strategies to improve adaptation efficiency.

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Data Availability All the datasets are publicly available.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tianqing reports financial support was provided by the National Natural Science Foundation of China under Grant 62676199 and 62176128. Tianqing reports financial support was provided by the Basic Research Program of Jiangsu under Grant BK20231143. Tianqing reports financial support was provided by the Open Project of Key Laboratory of Tibetan Information Processing of the Ministry of Education under Grant QHSF-CS-2609. Tianqing reports financial support was provided by the Fundamental Research Funds for the Central Universities No. NJ2023032. Tianqing reports financial support was provided by the 333 High-Level Talent Project of Jiangsu Province. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors and Affiliations

Qing Tian^{1,2,3} · Yongjiang Liu¹ · Keyang Cheng⁴ · Weihua Ou⁵ · Jianping Gou⁶

✉ Qing Tian
tianqing@nuist.edu.cn

Yongjiang Liu
liuyongjiang@nuist.edu.cn

Keyang Cheng
kycheng@ujs.edu.cn

Weihua Ou
ouweihua@gznu.edu.cn

Jianping Gou
cherish.gjp@gmail.com

- ¹ School of Computer Science, Nanjing University of Information Science and Technology, Nanjing 210044, China
- ² Wuxi Institute of Technology, Nanjing University of Information Science and Technology, Wuxi 214000, China
- ³ Key Laboratory of Tibetan Information Processing of the Ministry of Education, Qinghai Normal University, 810008 Xining, China
- ⁴ School of Computer Science and Communication Engineering, Jiangsu University, Zhenjiang 212013, China
- ⁵ School of Big Data and Computer Science, Guizhou Normal University, Guiyang 550025, China
- ⁶ College of Computer and Information Science and the College of Software, Southwest University, Chongqing 400715, China